

Sai Madhav

Full Stack Developer

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SUMMARY

- Having professional experience over 4 years as Full Stack Developer building and securing web applications across healthcare, automotive, and fintech domains in U.S based and global environments and strong experience in enterprise-scale application development and cloud-native architectures.
 - Expertise in Java, Spring Boot, ReactJS, Angular, GraphQL, Kafka, AWS, and modern DevOps practices. Proficient in implementing microservices, RESTful APIs, event-driven systems, and security protocols including OAuth2, JWT, and RBAC. Skilled in using design patterns like Singleton, Factory, and DAO for scalable and maintainable codebases.
 - Delivered a real-time stock analytics platform capable of processing 10K+ transactions per second using Kafka and ML models, significantly enhancing trading performance.
 - Developed and deployed a PCI DSS-compliant secure payment gateway with <200ms fraud detection response time, reducing transaction failures by 40% and boosting fraud detection accuracy by 30%.
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SKILLS

Java | Python | C/C++ | SQL | MySQL | PostgreSQL | MongoDB | DynamoDB | Redis | Apache Kafka | RabbitMQ | ReactJS | Angular | TypeScript | JavaScript | HTML5 | CSS3 | Node.js | Spring Boot | Hibernate | GraphQL | REST APIs | ISO 8583 | JSON | XML | AWS (EC2, S3, Lambda, DynamoDB, SQS, SNS, CloudWatch, SageMaker) | Azure | Jenkins | Docker | Kubernetes | Git | GitHub | GitLab | Bitbucket | Maven | CI/CD | ECS | EKS | OAuth2 | JWT | SSO | RBAC | Tokenization | API Encryption | HIPAA Compliance | PCI DSS Compliance | Fraud Detection | KYC/AML | Microservices | Micro Frontend | Event-Driven Architecture | Monolith to Microservices | Singleton | Factory | DAO | Circuit Breaker Patterns | Junit | Mockito | Jest | TestNG | Postman | Splunk | Log4j | IntelliJ IDEA | VS Code | Jira | Confluence | Agile | Scrum | TDD | BDD

EXPERIENCE

Full Stack Developer

CVS Health | Atlanta, GA

Aug 2023 - Present

Description: Developed cloud-native applications supporting patient health journeys, including medication management, benefits eligibility, and claim tracking using ML-enhanced insights. Delivered scalable and secure experiences for providers and members across CVS's digital ecosystem.

- Built responsive modules using ReactJS, Redux, and GraphQL to streamline pharmacy benefits workflows, enhancing system performance by 35% and user satisfaction by 30%.
- Implemented Micro Frontends for modular rollout of claims, prescriptions, and scheduling features, decreasing release dependency by 40%.
- Modernized legacy systems into Spring Boot microservices for real-time claim processing and eligibility checks, improving backend scalability by 45%.
- Leveraged GraphQL to power unified patient dashboards, reducing data fetch times by 50% and improving page load under heavy traffic.
- Developed a secure payment system for prescription copays using Java17, Kafka, ISO 8583, and ML-based fraud checks with sub-200ms latency.
- Enforced HIPAA compliance through OAuth2, JWT, RBAC, and Spring Security across all access layers.
- Automated deployment pipelines with Jenkins, Docker, GitHub Copilot, and AWS to reduce downtime and cut release cycles in half.

- Monitored services via AWS CloudWatch and Splunk, decreasing MTTR by 40% during prescription claim outages.
- Applied enterprise design patterns for scalable microservices managing PHI and PII data securely and efficiently.
- Coordinated with product, QA, and compliance teams across 4 major releases using Jira and Confluence to deliver on 95% sprint goals.

Environment: ReactJS, Redux, GraphQL, Java 17, Spring Boot, Spring Security, REST APIs, Kafka, ISO 8583, OAuth2, JWT, Jenkins, Docker, GitHub Copilot, AWS (EC2, Lambda, S3, CloudWatch), Splunk, MySQL, MongoDB, IntelliJ IDEA, Postman, Jira

Software Engineer

Mercedes-Benz Research and Development | Bengaluru, India

Jan 2022 – Jan 2023

Description: Led development of telematics, driver safety, and connected vehicle APIs for Mercedes-Benz's Digital Cockpit platform, enabling real-time data processing, alerts, and diagnostics for next-gen vehicles.

- Built Angular and React dashboards to visualize live vehicle telemetry, improving HMI response and usability by 40%.
- Migrated legacy vehicle systems to Spring Boot microservices on AWS for dynamic diagnostics and OTA updates.
- Developed Kafka pipelines to stream sensor events from vehicles to cloud backend, enabling 20% faster real-time alerts.
- Refactored safety-critical modules like braking and lane assist alerts with Spring Boot and DynamoDB, improving uptime during peak usage by 35%.
- Implemented OAuth2, JWT, and RBAC across connected mobility APIs, enforcing secure access and aligning with GDPR.
- Deployed and orchestrated Dockerized services in Kubernetes clusters, reducing infra provisioning effort by 45%.
- Applied resilience patterns (circuit breakers, fallbacks) in mission-critical telematics systems to ensure uptime under unstable networks.
- Supported QA, production support, and log debugging during test drives, cutting triage time by 60%.

Environment: Java/Spring Boot, Angular, ReactJS, Kafka, Docker, Kubernetes, AWS (EC2, S3, Lambda, DynamoDB), OAuth2, JWT, RBAC, Jenkins, Git, IntelliJ IDEA, Postman, Jira

Application Developer

Accenture | Hyderabad, India

Jan 2021 - Dec 2021

Description: Delivered client solutions in digital transformation projects using Java/J2EE and Azure cloud. Supported banking and energy-sector clients with UI modernization and middleware integrations.

- Built customer onboarding and KYC interfaces using AngularJS, HTML5, and JavaScript for responsive and secure form processing.
- Developed RESTful web services for backend integrations with credit scoring and account APIs.
- Styled consistent and performant layouts using CSS and jQuery, cutting UI latency by 25%.
- Deployed and managed apps on Azure App Services with automated pipelines in CI/CD.
- Wrote unit tests and automated builds via Jenkins and TestNG, ensuring regression stability.
- Managed Git repos in Bitbucket, collaborated on tasks and bug fixes via JIRA across global teams.
- Tuned exception handlers and added retry mechanisms for backend service stability.

Environment: Java/J2EE, AngularJS, Azure, HTML, CSS, JavaScript, jQuery, TestNG, Jenkins, Bitbucket, Log4j, JIRA, Agile

PROJECTS

AI-Powered Risk Engine for Health & Insurance

Description: Developed during engagements at CVS Health and Accenture, this platform evaluates customer risk profiles for underwriting and eligibility using real-time data from medical records, transactions, and behavior signals.

Responsibilities / Achievements:

- Developed backend microservices using Spring Boot to handle data ingestion, scoring logic, and reporting.
- Built ReactJS + GraphQL dashboard for underwriters and case managers.
- Processed financial/EHR data streams via Kafka for continuous risk profiling.
- Integrated Python-based ML models through SageMaker APIs.
- Enforced compliance via OAuth2, JWT, RBAC, HIPAA, and PCI DSS.
- Containerized and deployed services using Jenkins and AWS ECS.
- Delivered <250ms API latency and improved decision throughput by 40%.

Secure Payment Gateway for Copay & Mobility

Description: Built as part of CVS prescription copay processing and Mercedes-Benz's connected mobility payments system. Combined ISO 8583-based secure transactions with ML-powered fraud detection in a PCI DSS-compliant setup.

- Built secure Spring Boot modules for payment handling.
- Integrated ISO 8583 protocols and real-time fraud analytics with Kafka.
- Enabled PCI DSS compliance through encryption, tokenization, and auth standards.
- Achieved <200ms fraud response time; reduced failure rates by 40%, improved detection accuracy 30%.

EDUCATION

University of Alabama at Birmingham

Master of Science, Computer Science

Vellore Institute of Technology

Bachelor of Technology, Computer Science Engineering